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1  """
2  Example 4.6 Multiclass Stochastic Gradient Descent (SDG) for the MNIST data set
3  Machine Learning for Engineering Problem Solving
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5  """
6
7  import IPython as IP
8  IP.get_ipython().run_line_magic('reset', '-sf')
9
10 import numpy as np
11 import matplotlib.pyplot as plt
12 import sklearn as sk
13 import time as time
14 from sklearn import linear_model
15 from sklearn import pipeline
16 from sklearn import datasets
17 from sklearn import multiclass
18
19 cc = plt.rcParams['axes.prop_cycle'].by_key()['color']
20 plt.close('all')
21
22 %% Load your data
23
24 # Fetch the MNIST dataset from openml
25 mnist = sk.datasets.fetch_openml('mnist_784', as_frame=False, parser='auto')
26 X = np.asarray(mnist['data']) # load the data
27 Y = np.asarray(mnist['target'], dtype=int) # load the target
28
29 # Split the data set up into a training and testing data set
30 X_train = X[0:60000,:]
31 X_test = X[60000:,:]
32 Y_train = Y[0:60000]
33 Y_test = Y[60000:]
34
35 %% Train a Multiclass Stochastic Gradient Descent classifiers
36
37 # SK learn has a Multiclass and multilabel module as sk.multiclass. You can use
38 # this module to do one-vs-the-rest or one-vs-one classification.
39
40 # here we test a one-vs-rest classifier that uses Stochastic Gradient Descent
41 tt_1 = time.time()
42 ovr_clf = sk.multiclass.OneVsRestClassifier(sk.linear_model.SGDClassifier())
43 ovr_clf.fit(X_train, Y_train)
44 print('One-vs-Rest took '+str(time.time()-tt_1)+' seconds to train and execute')
45
46 # here we test a one-vs-one classifier that uses Stochastic Gradient Descent
47 tt_1 = time.time()
48 ovo_clf = sk.multiclass.OneVsOneClassifier(sk.linear_model.SGDClassifier())
49 ovo_clf.fit(X_train, Y_train)
50 print('One-vs-one took '+str(time.time()-tt_1)+' seconds to train and execute')
51
52 # Moreover, Scikit-Learn detects when you try to use a binary classification algorithm
53 # for a multiclass classification task, and it automatically runs OvA (except for SVM
54 # classifiers for which it uses OvO).
55 tt_1 = time.time()
56 multi_sgd_clf = sk.linear_model.SGDClassifier()
57 multi_sgd_clf.fit(X_train, Y_train) # y_train, not y_train_5
58 print('SK learns automated selection (OvA) took '+str(time.time()-tt_1)+' seconds to
59 train and execute')

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